Arrays and Classes

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Topics list

1. Arrays of different types
2. Arrays of primitive types
3. Arrays of objects
   - String
   - Person

Let’s Look at arrays of different types

Arrays can store any type of data

Let’s look at some examples:

1. Array of primitives - **int**
2. Array of objects – **String**
3. Array of objects - **Product**

An array can store any type of data.

Primitive Types

```
int numbers[] = new int[10];

byte smallNumbers[] = new byte[4];

char characters[] = new char[26];
```

Object Types

```
String[] words = new String[4];

Person[]  friends = new Person[4];
```

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Structure of an int primitive array

```
int[]  numbers;
```

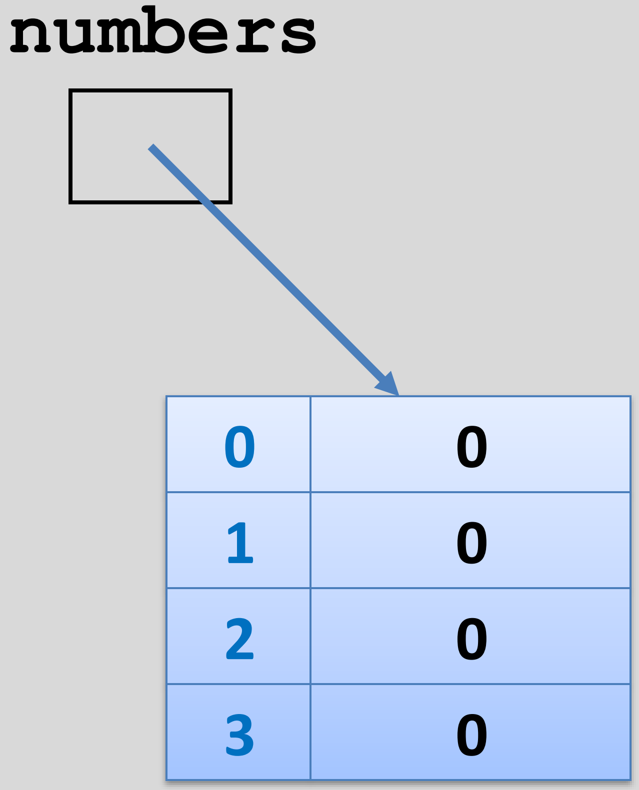
numbers

null

Structure of an **int** primitive array

```
int[]  numbers;
```

```
numbers = new int[4];
```



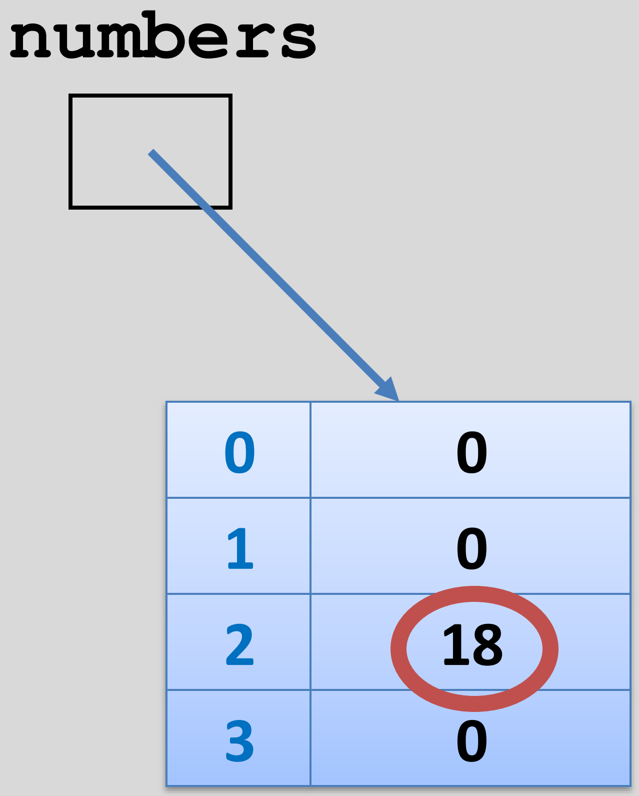
Structure of an **int** primitive array

int[] numbers;

numbers = new int[4];

numbers[2] = 18;

We are directly accessing the element at index **2** and setting it to a value of **18**.



Structure of an **int** primitive array

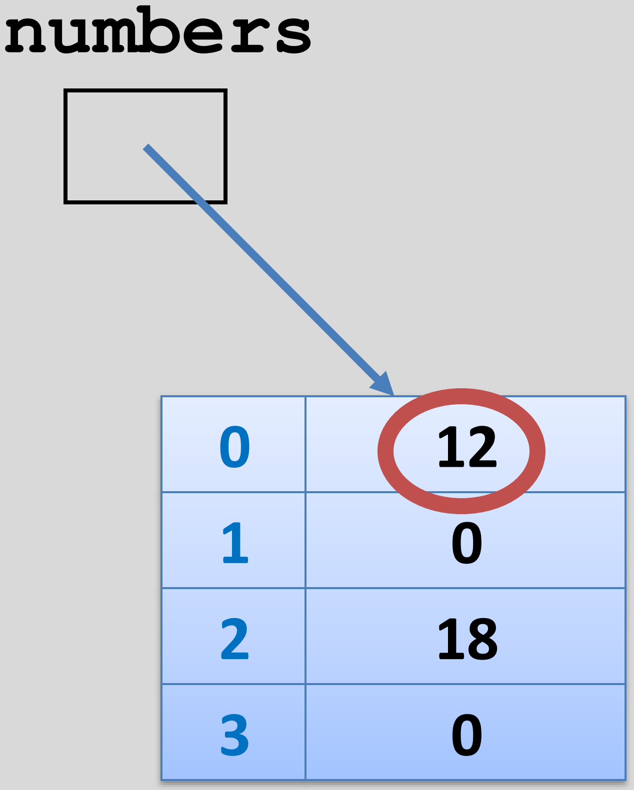
```
int[]  numbers;
```

```
numbers = new int[4];
```

```
numbers[2] = 18;
```

numbers[0] = 12;

We are setting the element at index **0** to a value of **12**.



Structure of an int primitive array

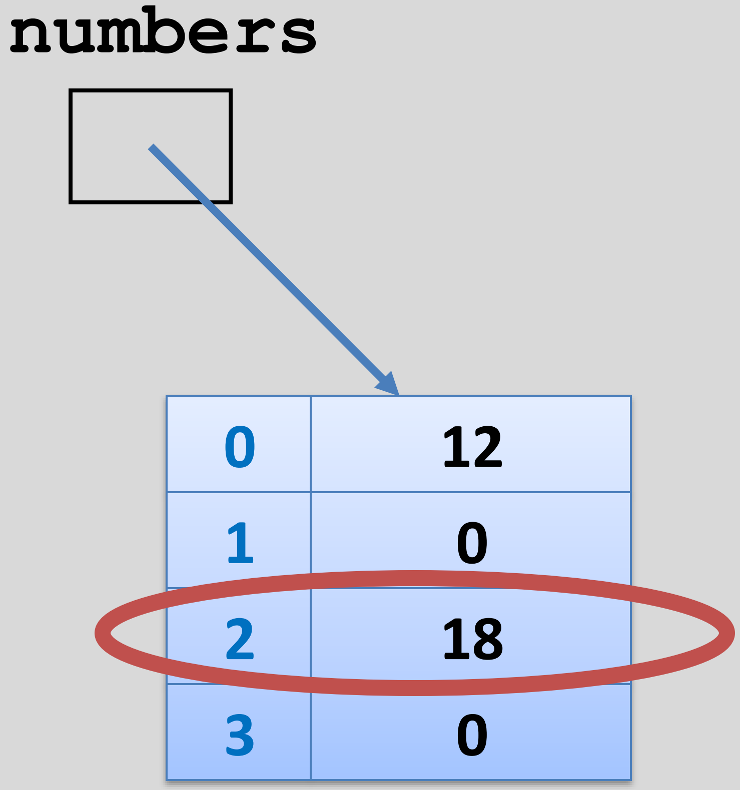
```
int[]  numbers;

numbers = new int[4];

numbers[2] = 18;

numbers[0] = 12;
```

```
print(numbers[2]);
```



Here we are printing the contents of index location 2
i.e. 18 will be printed to the console.

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Primitive Types

```
int numbers[] = new int[10];

byte smallNumbers[] = new byte[4];

char characters[] = new char[26];
```

Object Types

```
String[] words = new String[4];

Person[] friends = new Person[4];
```

Structure of a **String** object array

```
String[]  words;
```

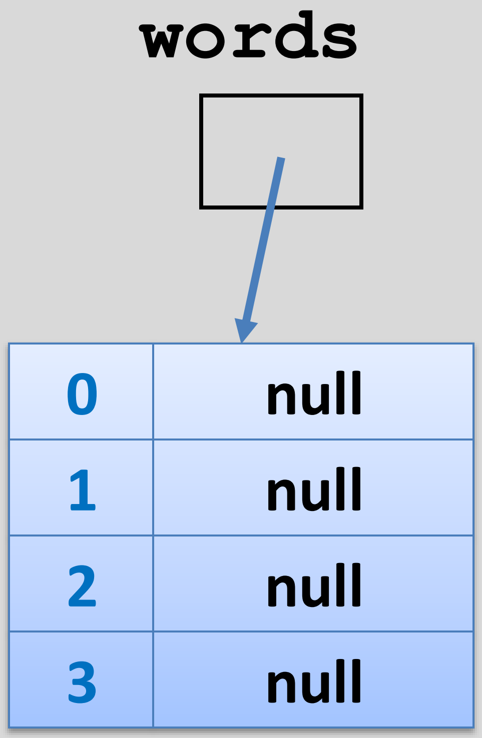
words

null

Structure of a String object array

```
String[]  words;
```

```
words = new String[4];
```

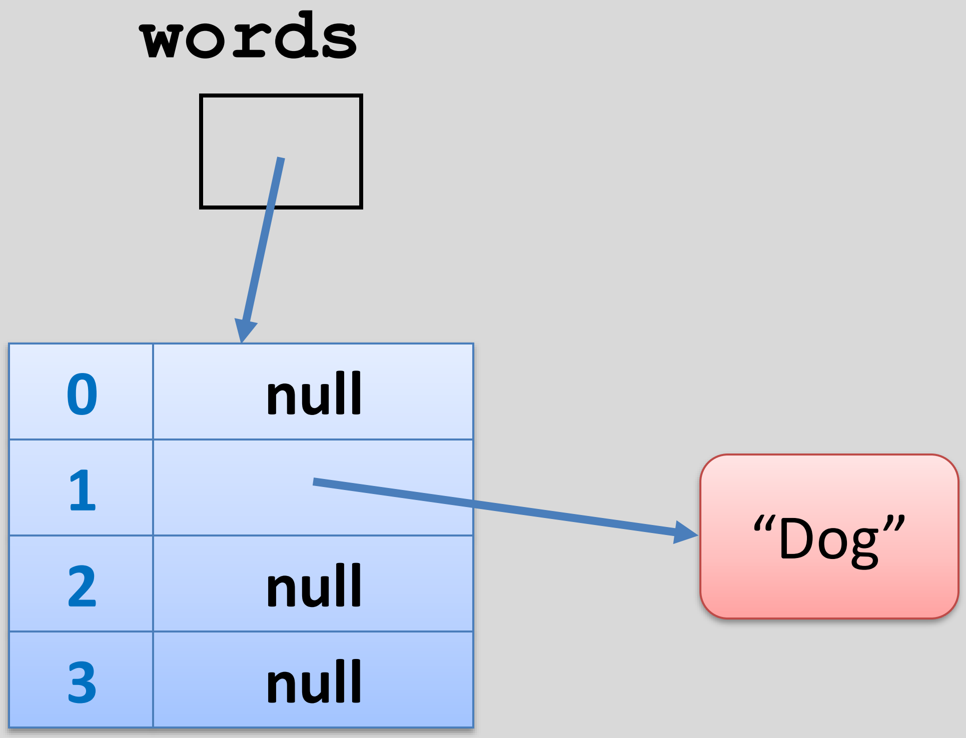


Structure of a **String** object array

```
String[]  words;
```

```
words = new String[4];
```

```
words[1] = “Dog”;
```



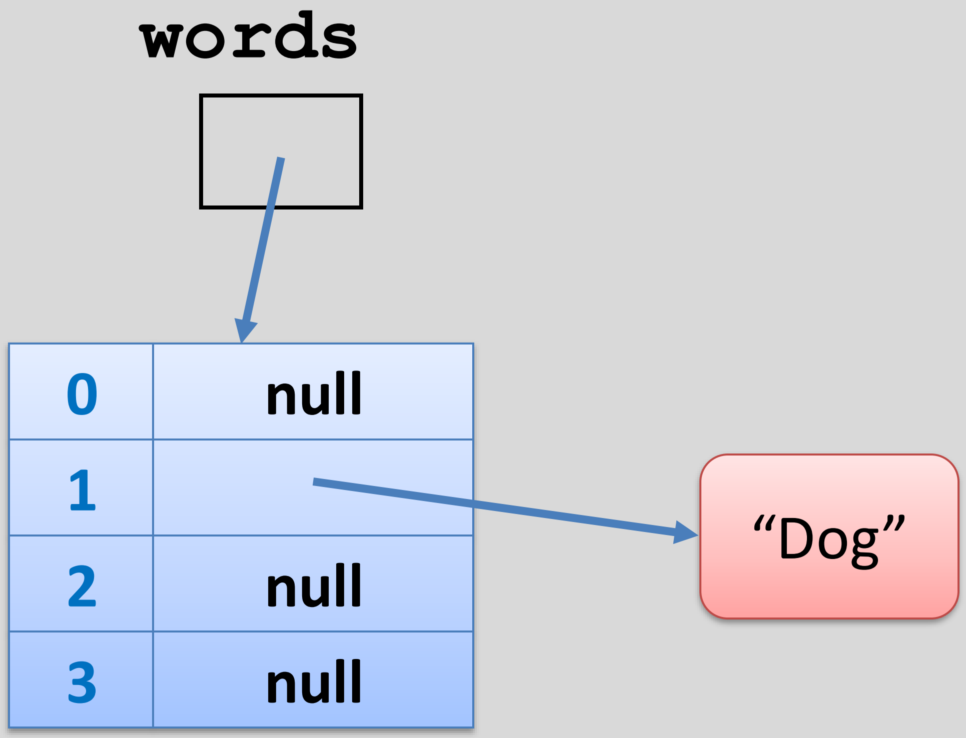
Structure of a **String** object array

String[] words;

words = new String[4];

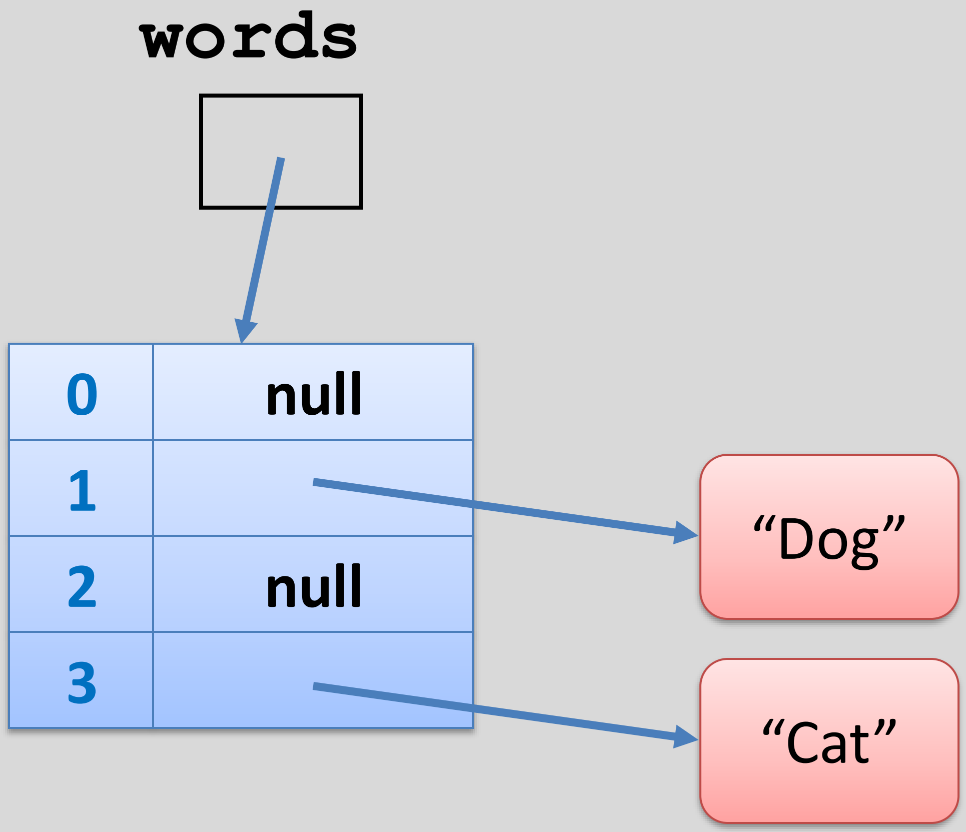
words[1] = “Dog”;

We are directly accessing the element at index **1** and setting it to a value of **“Dog”**.



Structure of a **String** object array

```
String[]  words;
words = new String[4];
words[1] = "Dog";
words[3] = "Cat";
```



Structure of a **String** object array

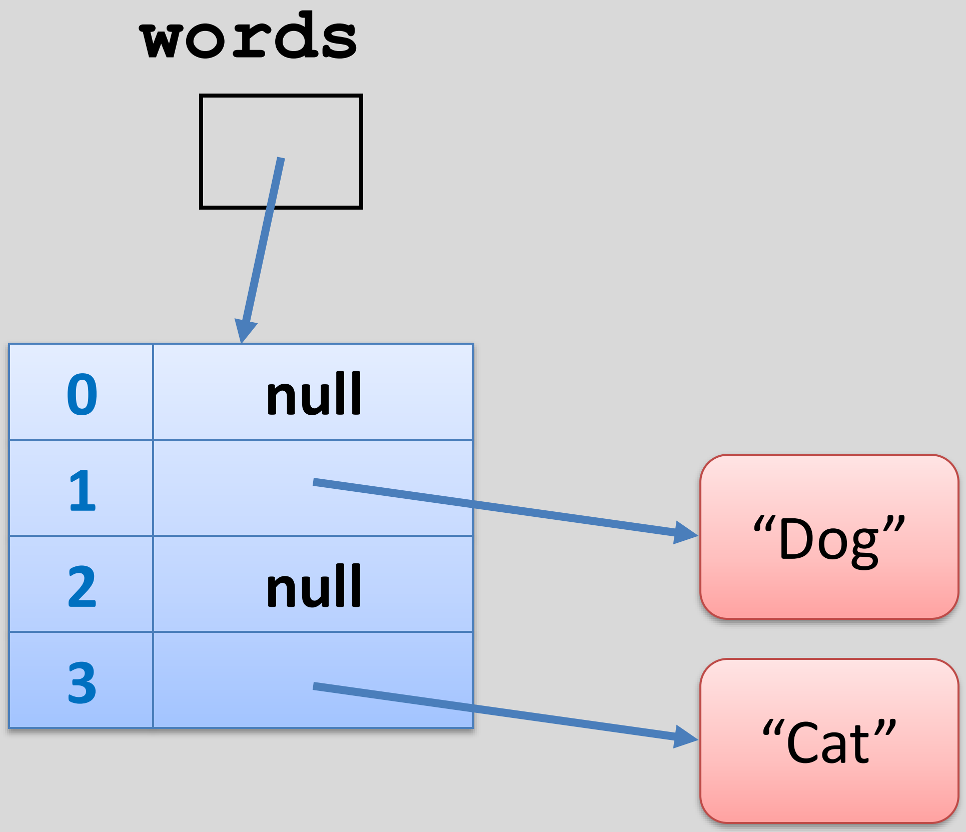
```
String[]  words;

words = new String[4];

words[1] = “Dog”;

words[3] = “Cat”;
```

The element at index **3** is set to **“Cat”**.



Structure of a **String** object array

```
String words[];

words = new String[4];

words[1] = "Dog";
words[3] = "Cat";

for (int i=0; i < words.length; i++)
{
   System.out.println(words[i]);
}
```

```
null
Dog
null
Cat
```

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An array can store any type of data.

Primitive Types

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int numbers[] = new int[10];

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Object Types

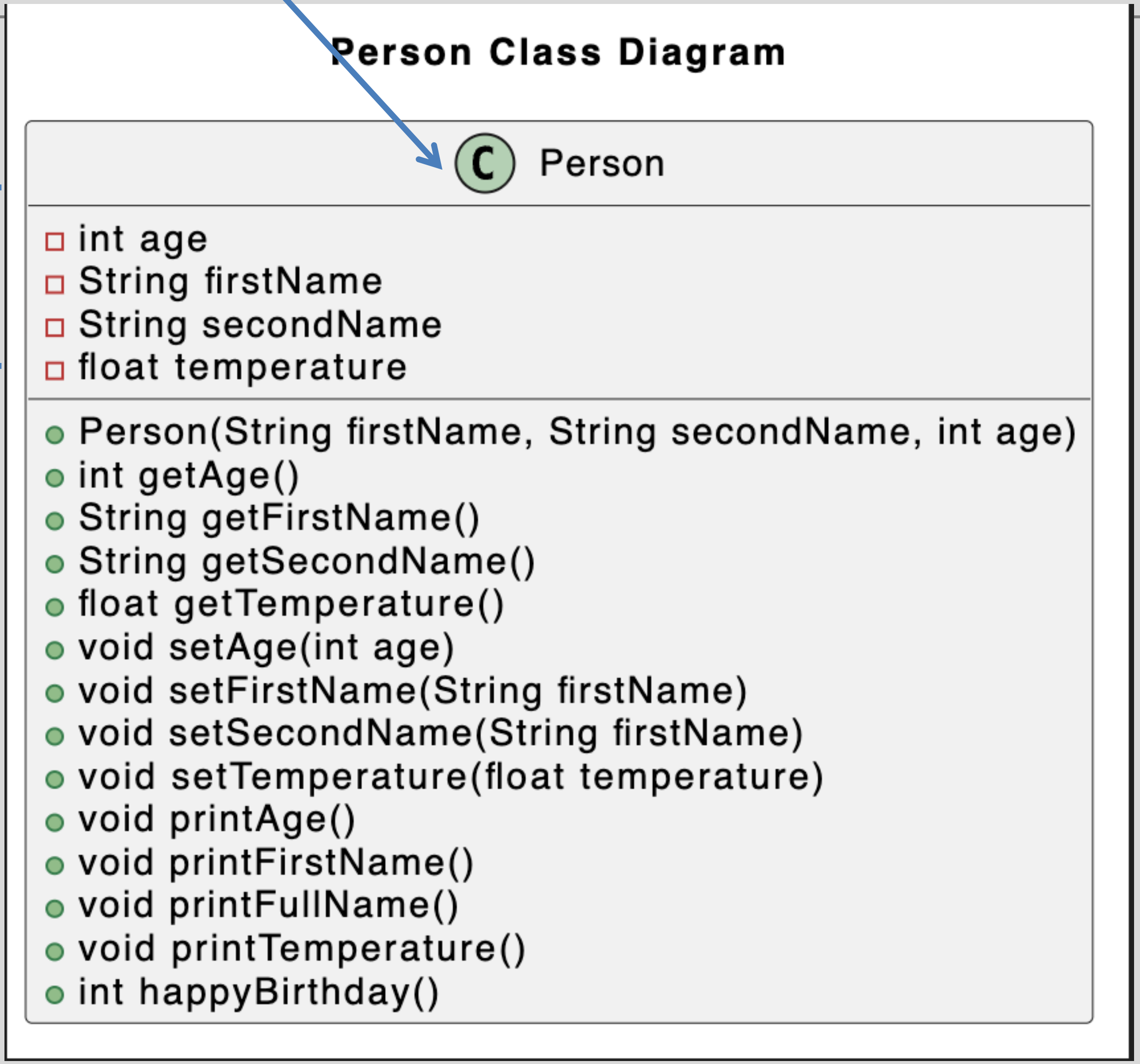
```
String[] words = new String[4];

  Person[] friends = new Person[4];
```

Person Class

Object Type/
Class Name

Fields
i.e. the attributes of the class

Methods
i.e. the behaviours of the class



Structure of a primitive array of **Person**

```
Person[]  friends;
```

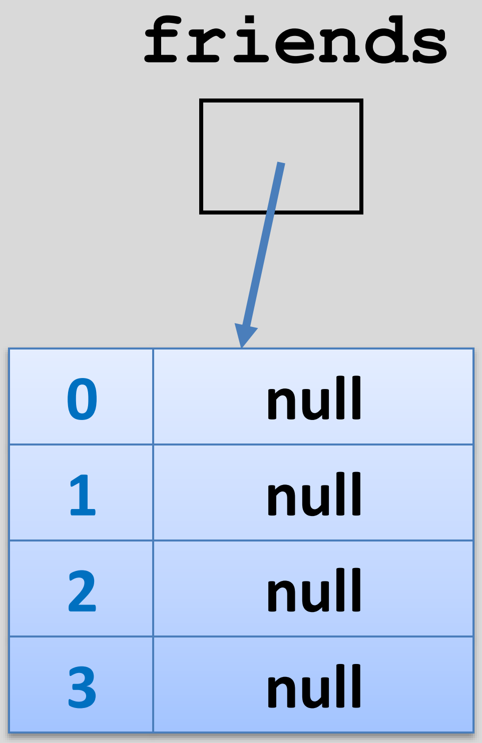
`friends`

null

Structure of a **Product** primitive array

```
Person[] friends;
```

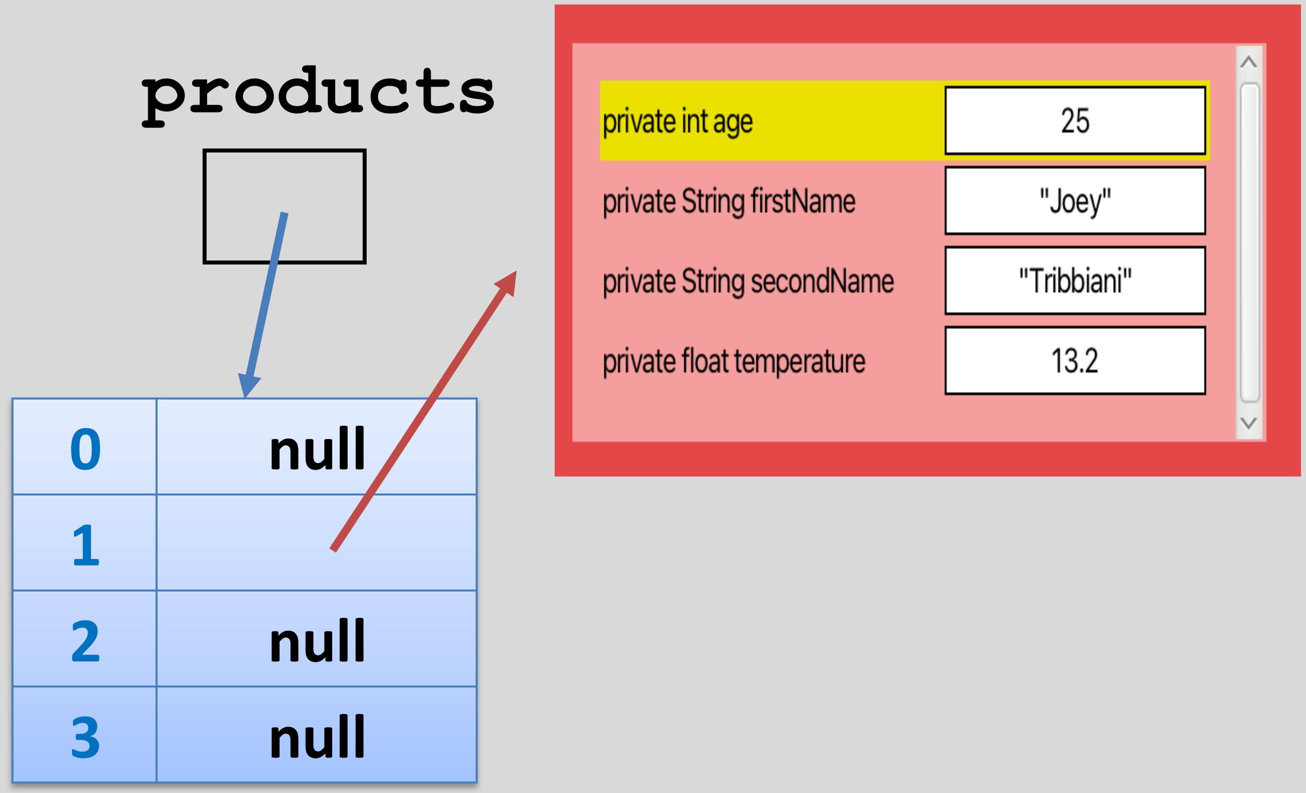
```
friends = new Person[4];
```



Structure of a **Product** primitive array

Person[] friends;

friends = new Person[4];



friends[1] = new Person("Joey", "Tribbiani", 25,);

Example using a **Person** object array

```
public String listFriends() {

      String listOfFriends = "";

      for (int i = 0; i < friends.length; i++) {
          friends[i]. printFirstName();
          friends[i]. printSecondName();
      }

      return listOfProducts;
   }
}
```

Returns the first and second names of all friends stored in the primitive array.

Questions?